

Cross-Lingual Dependency Parsing Report

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[Submission Date]

1 Introduction

1.1 Objective

In this experiment, we evaluate the impact of cross-lingual data augmentation on the performance of dependency parsing for Swedish (TGT). Specifically, we investigate the effectiveness of Norwegian (GTRF) and Chinese (CTRF) as transfer languages in improving dependency parsing performance.

1.2 Experiment Setup Overview

I conduct three experiments:

- Swedish Monolingual Parsing (baseline)
- Swedish + Norwegian (Good Transfer Language - GTRF)
- Swedish + Chinese (Contrastive Transfer Language - CTRF)

Evaluation Metrics:

- Unlabeled Attachment Score (UAS)
- Labeled Attachment Score (LAS)
- Weighted LAS

2 Language Selection

2.1 Target Language (TGT): Swedish

Swedish was chosen as the target language because it is a moderately resourced language in the Universal Dependencies (UD) dataset and is structurally similar to other Germanic languages, making it an interesting case to examine transfer learning.

2.2 Good Transfer Language (GTRF): Norwegian

Norwegian was chosen because it is extremely close to Swedish. They both belong to the North Germanic branch of the Indo-European language family, with similar word order, morphological structure, and lexical similarities.

2.3 Contrastive Transfer Language (CTRF): Chinese

Chinese was selected as the contrastive transfer language as it is typologically distant from Swedish. Chinese, unlike Swedish, is an analytic language with little inflection, differing significantly in syntax, and an SVO word order without the rich morphological cues of Swedish.

3 Data Preparation and Experiments

Data is taken from Universal Dependencies (UD) v2.15, located in:
/common/student/corpora/ud-treebanks-v2.15

Language	Treebank	Training Sentences	Development Sentences
Swedish (TGT)	UD_Swedish	100	100
Norwegian (GTRF)	UD_Norwegian	500	100
Chinese (CTRF)	UD_Chinese	500	100

3.1 Monolingual Swedish Model

This setup is designed to train a parser based solely on Swedish data. The purpose is to evaluate the model’s performance in a single-language setting.

3.2 Multilingual Swedish + Norwegian Model

This model is trained using both Swedish and Norwegian data. The objective is to evaluate the parser’s performance when exposed to multiple languages, particularly those within the same language family.

3.3 Multilingual Swedish + Chinese Model

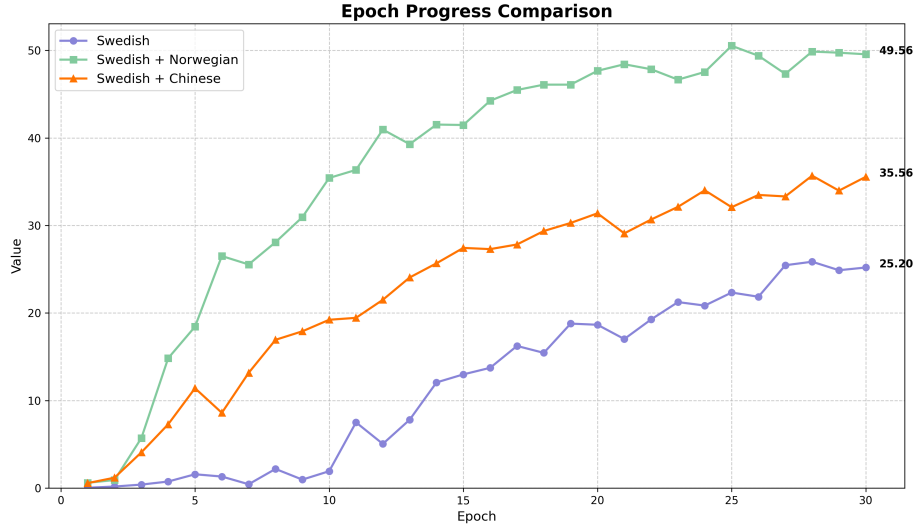
In this setup, the parser is trained on both Chinese and Swedish data. The purpose is to test the parser’s ability to handle languages from distinct language families, requiring it to accommodate a greater diversity of linguistic forms.

4 Results and Analysis

4.1 UAS & LAS Scores

Model	UAS	LAS	Weighted LAS
Swedish Monolingual	44.64	25.86	19.31
Swedish + Chinese	48.90	35.56	29.46
Swedish + Norwegian	61.06	49.56	43.16

4.2 Learning Curves



4.3 Analysis

1. Green Curve: Swedish + Norwegian

The green curve (Swedish + Norwegian) has the highest growth rate, indicating that the model effectively learns features from the very beginning. The performance stabilizes at around 25-30 epochs, with a final LAS of 49.56. This demonstrates that Swedish and Norwegian languages share significant linguistic knowledge, allowing for good knowledge transfer and substantially improving parsing performance.

2. Orange Curve: Swedish + Chinese

The orange curve (Swedish + Chinese) demonstrates moderate improvement but remains lower than the Swedish + Norwegian model, with a final LAS of 35.56. This indicates that adding Chinese data provides some benefit but is less effective due to the structural differences between the two languages.

3. Blue Curve: Swedish Monolingual

The blue curve (Swedish Monolingual) grows the slowest, with a final LAS of only 25.20. This suggests that a single-language model struggles to generalize as effectively as multilingual approaches.

5 Conclusions and Discussion

1. Cross-lingual Transfer Effect

From the learning curves, the Swedish + Norwegian model performs the best. This can be attributed to the strong linguistic similarities between the two languages. Since Swedish and Norwegian belong to the North Germanic language family, they share syntax, morphology, and vocabulary. As a result, the model effectively transfers useful features from Norwegian to improve Swedish parsing performance.

2. Impact of Linguistic Distance

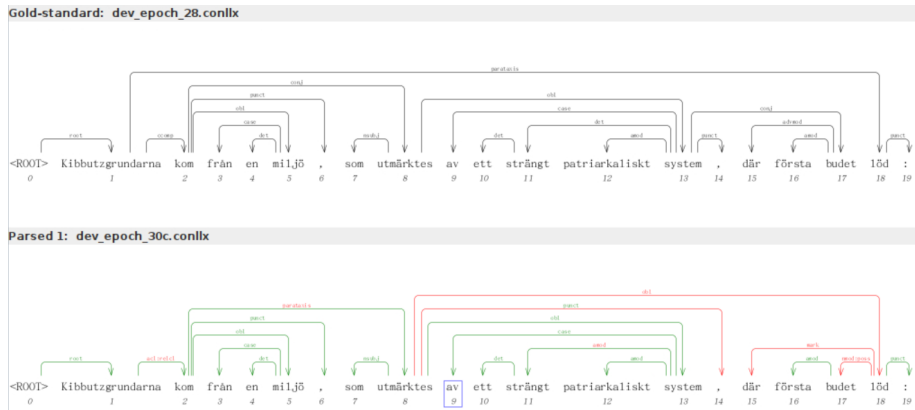
In contrast, the Swedish + Chinese combination shows only a moderate improvement. This is likely due to the structural differences between Swedish and Chinese, as they belong to different language families (Indo-European vs. Sino-Tibetan). Differences in word order, syntax, morphology, and grammatical rules make it more challenging for the model to transfer knowledge effectively. This highlights that language similarity is a key factor in successful cross-lingual learning. For linguistically distant pairs like Swedish and Chinese, additional techniques (such as subword representations, shared encoders, or multilingual pretraining) may be necessary to enhance transferability.

3. Importance of Data Diversity

The Swedish monolingual model performs the worst, emphasizing that a model trained on a single language lacks diversity and struggles to generalize. Adding multilingual data, particularly from structurally similar languages, significantly enhances model performance by introducing more varied syntactic patterns. This aligns with the general assumption in multilingual learning that diverse data sources improve generalization.

5.1 Qualitative Error Analysis

5.1.1 example1



The qualitative evaluation compares dependency parsing errors between the gold-standard parse and the system-generated parse. The gold-standard rep-

resents the correct parsing structure, while the parsed output from the system contains errors highlighted in red, with correctly parsed dependencies shown in green.

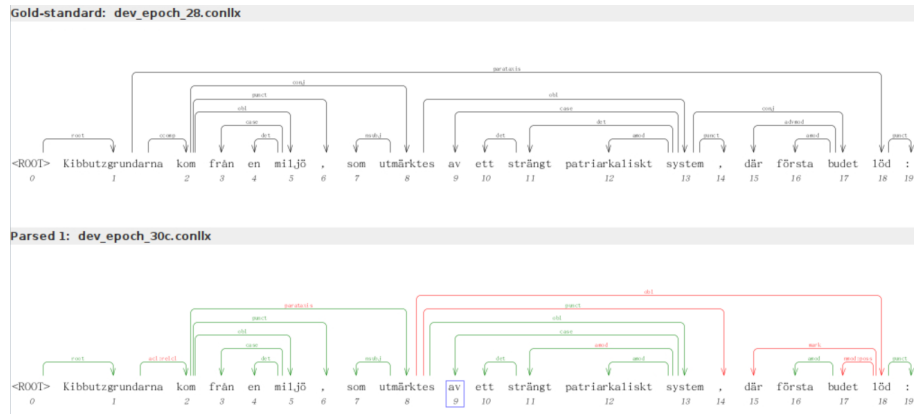
One notable difference is the attachment of the word "där," which in the gold-standard correctly modifies "löd," meaning "sounded." In the system output, "där" is incorrectly attached to "system." This misattachment may be due to the model's difficulty in handling relative adverbs in similar sentence structures, potentially because of insufficient training examples.

Another issue is the classification of "som." In the gold-standard, "som" correctly serves as the subject of "utmärktes," meaning "was characterized." However, in the parsed output, it is incorrectly linked to "kom" and labeled as a relative clause modifier. This suggests that the model struggles with parsing Swedish relative clauses, possibly misinterpreting "som" as modifying the main verb instead of forming a relative clause.

There is also an error in the classification of "första." In the gold-standard, "första" correctly modifies "budet" as an adjectival modifier, while in the system output, it is misclassified as a subordination marker. This confusion could be due to an imbalance in training data, with the model lacking sufficient examples to distinguish between adjectival modifiers and subordination markers.

Overall, the errors primarily involve relative clause structures, adverbial attachment, and modifier classification. The system struggles with complex dependencies, particularly in distinguishing relative clauses and modifier relationships. Improving the model could involve increasing training data, refining feature extraction methods, and using multilingual models to enhance generalization across different linguistic structures.

5.1.2 example2



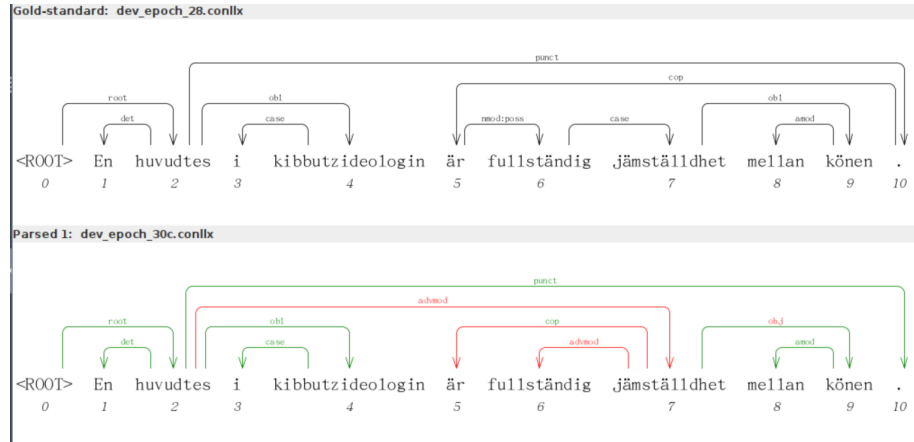
One key difference is the attachment of the phrase "I den traditionella familjen" ("In the traditional family"). In the gold-standard, "I" is correctly labeled as a case marker for "familjen," forming a prepositional phrase that modifies the main verb "grundades" ("was founded"). However, in the system output, "I" is incorrectly attached directly to the root, suggesting a misinterpretation

of its syntactic role. This may indicate a weakness in handling prepositional phrases, possibly due to an imbalance in training data.

Another error involves the subject "fadersauktoriteten" ("the paternal authority"). In the gold-standard, it is correctly labeled as a passive subject (nsubj:pass), indicating that the action "grundades" ("was founded") applies to it. In the system output, however, it is misclassified as an adverbial modifier, which alters the sentence's meaning. This suggests that the model struggles with correctly identifying passive constructions, which may be due to insufficient exposure to similar sentence patterns.

The model also makes a mistake in classifying "traditionella" ("traditional"). In the gold-standard, it is correctly linked to "familjen" ("the family") as an adjectival modifier, while in the system output, it is misclassified as an amod dependency directly connected to "grundades." This error could stem from the model's difficulty in distinguishing between adjectives modifying nouns and adjectives incorrectly linked to verbs.

5.1.3 example3



One key difference in the sentence involves the attachment of the verb "är" ("is"). In the gold-standard parse, "är" is correctly labeled as a copula (cop) linking "fullständig" ("complete") to "jämställdhet" ("equality"). However, in the system output, "är" is instead classified as an adverbial modifier (advmod), which misrepresents its grammatical function. This suggests that the model may have difficulty distinguishing between copular constructions and adverbial dependencies.

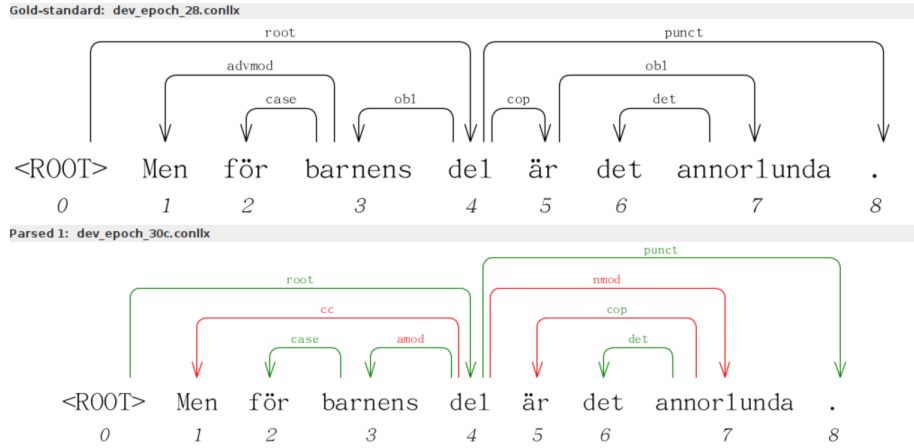
Another major error is the attachment of the word "huvudtes" ("main thesis"). In the gold-standard parse, it is correctly identified as the root of the sentence, serving as the subject of the copular construction. However, in the system output, "huvudtes" is incorrectly attached to "är", suggesting a misunderstanding of sentence structure. This misclassification might indicate a weakness in correctly identifying nominal predicates in copular sentences.

Additionally, the relationship between "mellan" ("between") and "könen"

("genders") is handled differently in both parses. In the gold-standard, "mellan" is labeled as a case marker for "könen", properly forming a prepositional phrase. In contrast, the system output incorrectly labels "mellan" as an object (obj), which distorts the meaning of the sentence. This error highlights the system's struggle with correctly interpreting prepositional phrase structures.

These errors suggest that the model might struggle with handling copular constructions, root identification, and prepositional phrase attachments, possibly due to imbalanced training data or insufficient exposure to similar sentence patterns.

5.1.4 example4



One key difference in this sentence is the attachment of "Men" ("But"). In the gold-standard parse, it is correctly classified as an adverbial modifier (advmod) linked to the root verb "är" ("is"). However, in the system output, it is incorrectly treated as a coordinating conjunction (cc) and attached directly to the root. This suggests that the model misinterpreted "Men" as linking clauses rather than modifying the main clause.

Another significant error is the treatment of "barnens" ("children's"). In the gold-standard, it is correctly labeled as a possessive modifier (case) of "del" ("part"), forming a noun phrase. However, in the system output, "barnens" is misclassified as an adjectival modifier (amod) of "del", which is incorrect. This may indicate that the model struggles with possessive constructions, possibly due to inadequate exposure to similar syntactic structures.

Additionally, the phrase "för barnens del" ("for the children's part") is parsed differently across systems. In the gold-standard, "för" is marked as a case marker for "barnens del", correctly forming a prepositional phrase. In contrast, in the system output, "del" is treated as the head of a nominal modifier (nmod) structure, incorrectly implying a different syntactic relationship. This suggests that the model has difficulty properly handling prepositional phrases and their dependencies.

Lastly, the model correctly identifies "är" as a copula (cop) linking "det"

("it") to "annorlunda" ("different"). However, due to errors in earlier phrase attachments, the overall syntactic structure is misrepresented.

These errors highlight potential weaknesses in handling adverbial modifiers, possessive constructions, and prepositional phrases, which could be due to insufficient training data or bias toward specific syntactic patterns.